

Supporting Figures for
Simple Code, More Scrutiny:
Python Proficiency and Review Effort in AI Agent and
Human Pull Requests

Chaiyong Ragkhitwetsagul, Nanthit Temkulkiat, Morakot Choetkiertikul,
Ruksit Rojpaisarnkit, Raula Gaikovina Kula, and Matheus Paixao

This document contains supporting figures referenced from the main article and distributed with its replication package. They present the same underlying data as tables that appear in the article, at a finer granularity, and are provided here so that the article itself remains within its page budget.

1 Per-agent proficiency by PR task type

Figures S1 and S2 split each PR task type further by agent, showing the proficiency distribution of every task type and agent combination as a separate panel. Figure S1 covers the five task types with the most PRs, and Figure S2 the remaining five. They expand the aggregated task-type breakdown given in Table 4 of the article.

A panel is left empty when the agent has no merged PR of that task type in our sample. This is the case for build PRs by Devin and Cursor, CI PRs by Copilot and Cursor, performance PRs by Devin, and PRs labelled `other` by Copilot and Devin. The panels confirm that the dominance of A1 constructs holds for every agent within every task type, and that the differences between task types are larger than the differences between agents within a task type.

The sparsely populated panels should not be over-interpreted. The docs panel for Cursor, for instance, consists of a single PR contributing only two constructs, both at B1, which is why it departs from the pattern shown by the other documentation PRs. The PR counts behind every panel are listed in the accompanying data files.

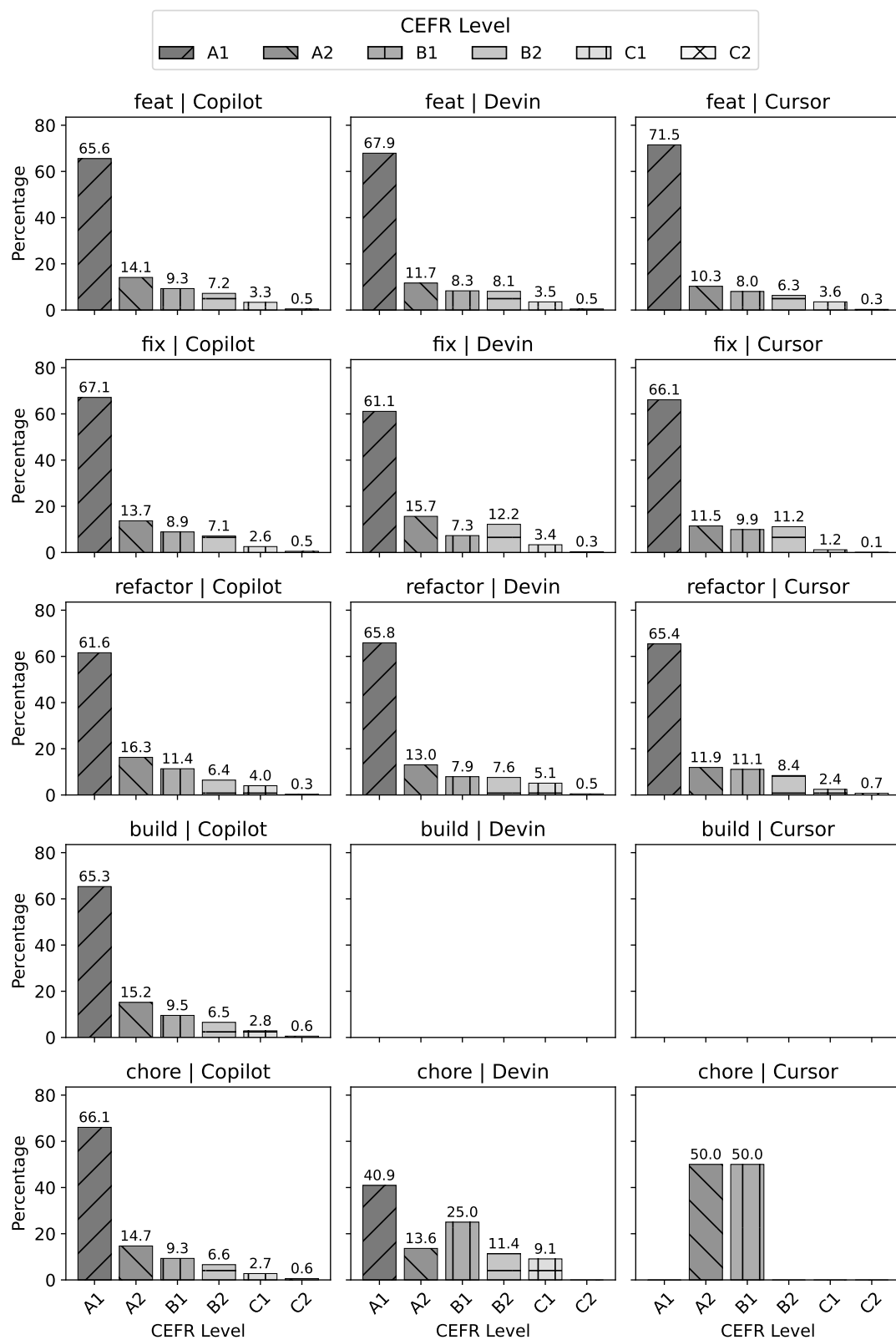


Figure S1: Distribution of code proficiency generated by AI agents for the 5 task types: feat, fix, refactor, build, and chore

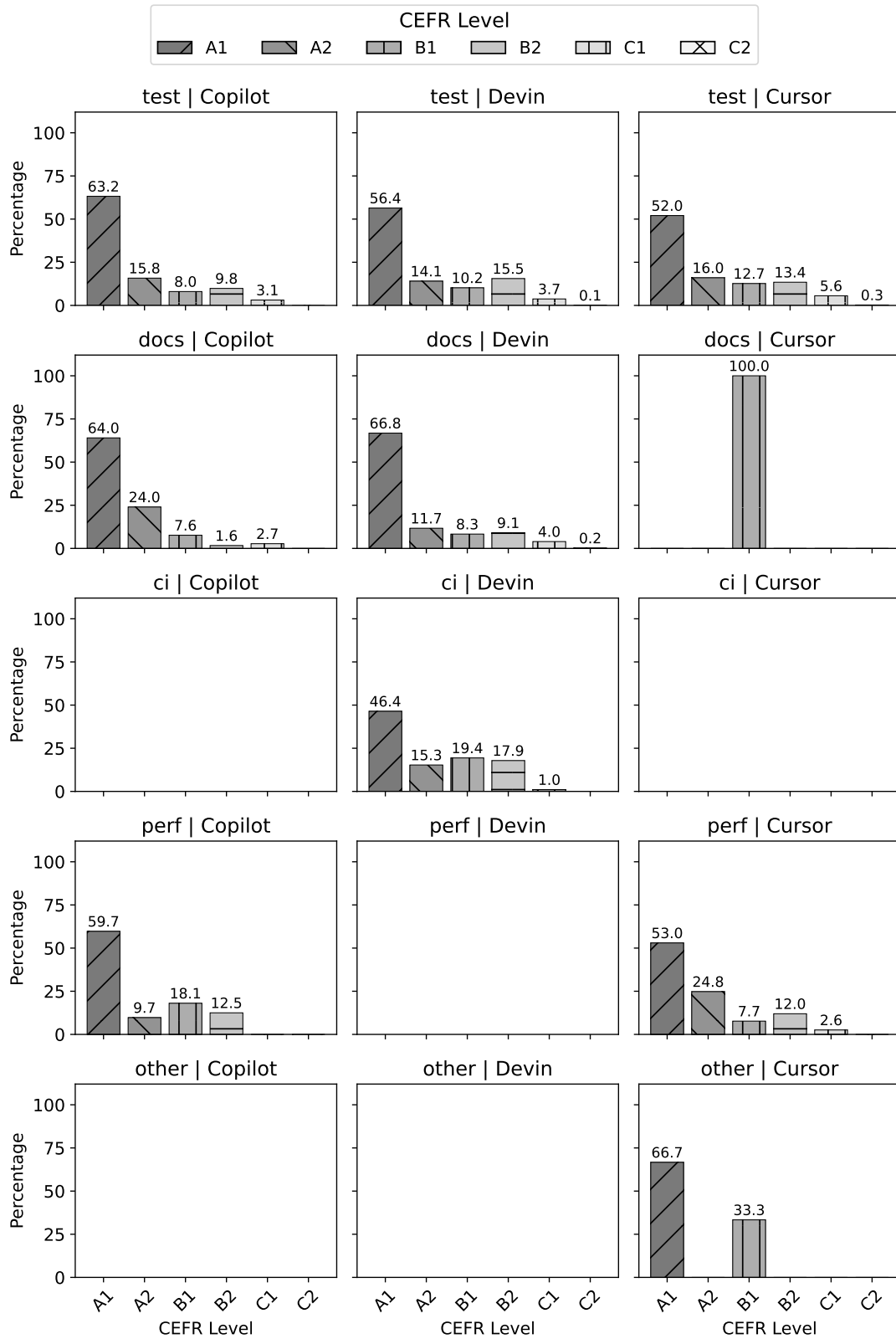


Figure S2: Distribution of code proficiency generated by AI agents for the remaining task types: test, docs, ci, perf, and other

2 Per-repository proficiency across the three groups

Figure S3 shows the mean ordinal CEFR score of each group in each of the twelve repositories to which all three groups contributed code. It accompanies the repository-paired Wilcoxon signed-rank tests reported in RQ2 of the article. The three groups exchange positions from one repository to the next, and no group is systematically higher than the others, which is consistent with the tests finding no difference for any pair.

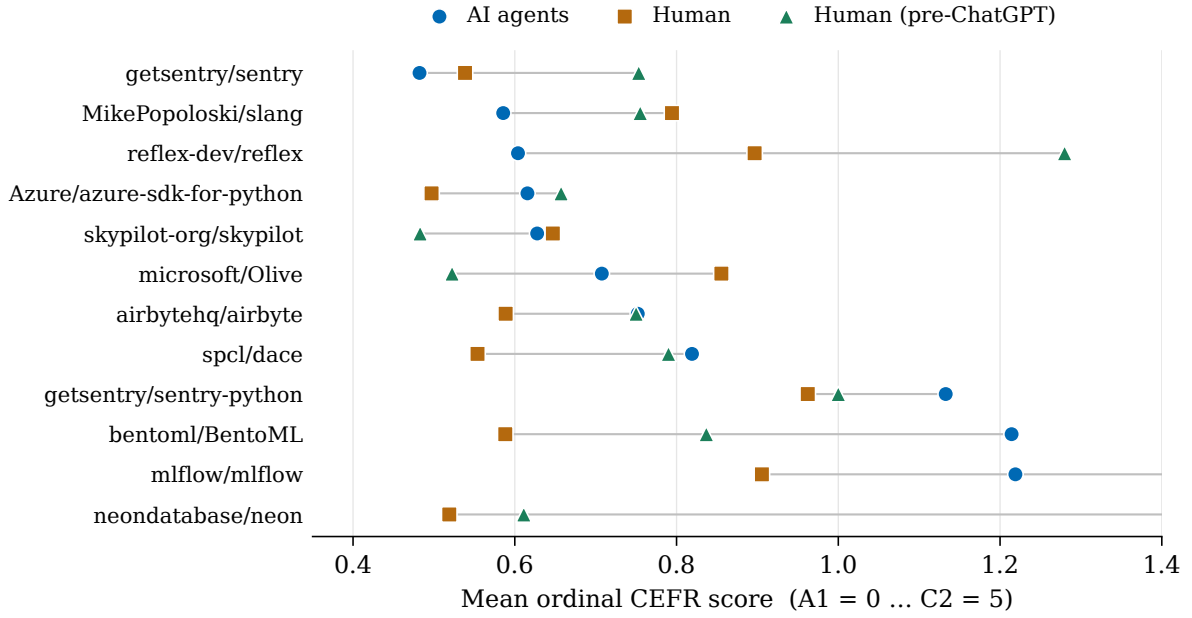


Figure S3: Mean ordinal CEFR score per repository for the three groups, across the twelve repositories present in all three. Repositories are ordered by the AI agents' score.